

**WEATHER: sMAPE at each forecast step**  
**Lookback = 168 steps, Forecast horizon = 12 steps**

| Forecast step | CNN    | LSTM   | dCeNN+ELM | dCeNN+ELM+ASP |
|---------------|--------|--------|-----------|---------------|
| 1.000         | 59.179 | 61.762 | 72.590    | 51.977        |
| 2.000         | 61.342 | 62.008 | 72.649    | 52.177        |
| 3.000         | 62.816 | 63.038 | 72.627    | 52.151        |
| 4.000         | 63.699 | 63.797 | 72.565    | 52.151        |
| 5.000         | 64.486 | 64.517 | 72.578    | 52.447        |
| 6.000         | 65.333 | 65.164 | 72.594    | 52.573        |
| 7.000         | 65.800 | 65.743 | 72.622    | 52.534        |
| 8.000         | 66.295 | 66.234 | 72.753    | 52.831        |
| 9.000         | 66.455 | 66.660 | 72.846    | 53.224        |
| 10.000        | 66.945 | 67.079 | 72.786    | 53.204        |
| 11.000        | 66.911 | 67.423 | 72.728    | 53.335        |
| 12.000        | 67.298 | 67.875 | 72.660    | 53.172        |

- This table shows how prediction error changes from the first forecast step to the last forecast step.
- It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.
- Lower is better. This is a percentage-style error that is easier to compare across scales.
- Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.